Cyber-Physical Systems (CPS) Smart Honeypot: System Architecture & Specifications

Project Name: Smart Interactive Honeypot for Cyber-Physical Systems

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1. System Overview

The Cyber-Physical Systems (CPS) Smart Honeypot is a lightweight, multi-protocol security system engineered to safely capture, log, and analyze threat actor interactions targeting industrial control networks and IoT infrastructure. The platform simulates standard industrial and IoT protocols (Modbus/TCP, MQTT, HTTP SCADA, SSH) while feeding real-time interaction payloads into a machine learning classification engine mapped against the **MITRE ATT&CK for ICS** framework.

2. Layered Architecture Breakdown

2.1 Emulation Layer (Interactive Socket Servers)

- **Modbus/TCP Emulator (Port 5020):** Listens for industrial protocol manipulation, function code execution, and holding register read/write commands. Responds with standard Modbus exception frames (0x83) to mimic operational hardware.
- **MQTT Broker Emulator (Port 1883):** Handles Industrial IoT message broker connection handshakes (CONNECT) and telemetry publication attempts, returning valid CONNACK frames.
- **HTTP SCADA Portal Emulator (Port 8080):** Emulates a Human-Machine Interface (HMI) control panel vulnerable to web vulnerability scans, SQL injection, and command injection attacks.
- **SSH Gateway Emulator (Port 2222):** Captures automated brute-force login attempts and initial access probes targeting secure industrial management terminals.

2.2 Logging & Storage Layer

- Intercepts network socket data and metadata: standard ISO timestamps, source IP, source port, targeted protocol service, and raw request payloads.
- Serializes log entries into a standardized, structured JSON store

(02_Honeypot_Code_and_Logs/honeypot_attacks.json) for downstream ingestion.

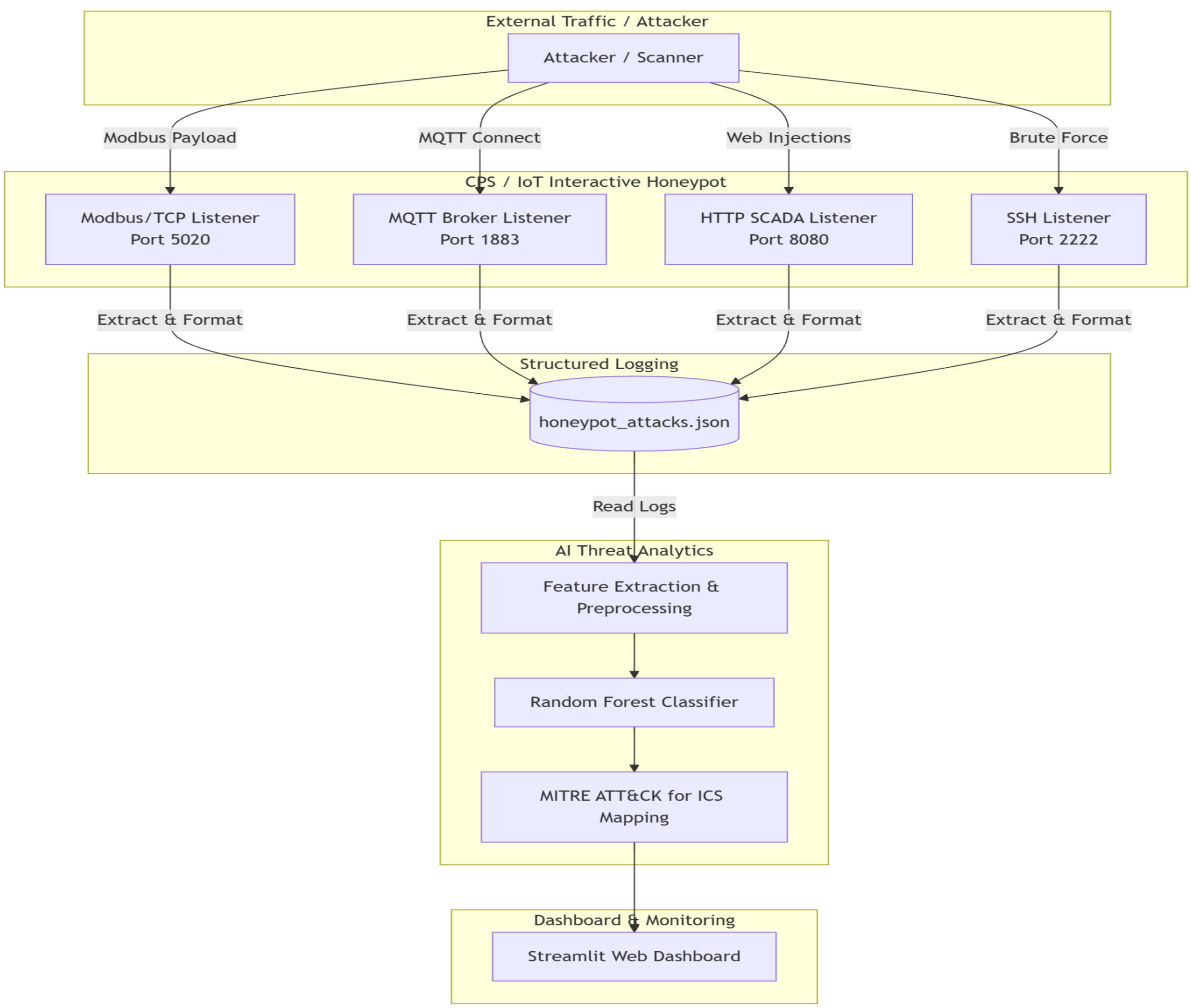
2.3 Analytics & Threat Classification Layer

- **Feature Extraction:** Extracts numerical features including payload length, interaction character distributions, and one-hot encoded target services.
- **Machine Learning Classifier:** Executes anomaly detection and threat vector classification using a Random Forest algorithm.
- **Taxonomy Mapping:** Cross-references classified attack behaviors directly with standard MITRE ATT&CK for ICS technique identifiers (e.g., T0855, T0886, T0812, T0847).

2.4 Visualization Layer

- Interactive web monitoring dashboard built using **Streamlit** (app.py).
- Displays high-level event metrics, target protocol distribution, attack vector taxonomy, and searchable real-time incident telemetry tables.

3. High-Level Data Flow Diagram



4. Hardware & System Resource Profile

Specification	Minimum Requirement	Lightweight Host Optimization
CPU	Dual-core 1.5 GHz	Dual-core 2.0 GHz
RAM Utilization	< 500 MB	< 250 MB Active Footprint
Storage Space	< 50 MB	< 15 MB Total Directory Size
Runtime Environment	Python 3.8+	Python 3.10+
Network Interface	Bound to local interface (0.0.0.0)	Isolated Virtual Socket Layer